Unit - 3 COA

Q. write the differences between RISC and CISC.

Ans

RISC	CISC
1. Highly Pipelined	1. Less pipelined.
2. Fixed length instructions.	2. Variable length instructions.
3. Few addressing mode.	3. Many addressing modes.

Q. what are types of microinstructions available?

Ans Following types of microinstructions are available?

i) Vertical / Horizontal
ii) Packed / unpacked.
iii) Hard / soft microprogramming
iv) Direct / indirect encoding.

Q. What is parallelism and pipelining in Computer architecture?

Ans i) Parallelism in Computer architecture:

1. Executing two or more operations at the same time is known as parallism.
2. The purpose of parallel processing is to speed up the Computer processing capblity.
3. It also increases throughout, i.e amount of processing that can be accomplished during a given interval of time.
4. Improves the performance of the computer for a given clock speed.
5. Two or more ALUs in CPU can work concurrently to increase throughput. The system may have two or more processors operating concurrently.

ii) Pipelinging:

1. Pipelining is a technique of decomposing a sequential process into sub-operations, with each sub-process being executed in a special dedicated segment that operates concurrently with all other segment.
2. The processor executes a program by fetching and executing instructions, one after the other.

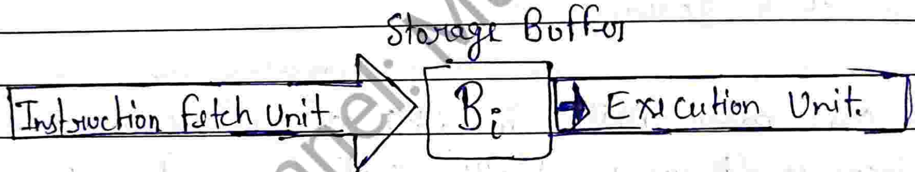
I_1		I_2		I_3		I_4	
F_1	E_1	F_2	E_2	F_3	E_3	F_4	E_4

3. Now consider a computer that has two separate hardware units, one for fetching instructions and another for executing them, as shown.

4. The instruction fetched by the Fetch unit is deposited in an intermediated stroge buffer B_i.

5. The results of execution are deposit in the destination location specified by the instruction.

6. For these purposes, we assume that both the source and destination of the data operated in by the instructions are inside the block labelled "Execution unit".



7. The computer is controlled by a clock whose period is such that the fetch and execute steps of any instruction can be completed in one clock ~~cycles~~ cycle.

Q. Explain the Organization of microprogrammed Control unit in detail.

Ans A microprogrammed control unit (MCU) is a type of control unit used in microprocessors, which uses a microprogram to control the sequence of operation in the ~~program~~ processor. The organization of a microprogrammed control unit consists of the following components:

1. Control Memory : The control memory contains the microprogram that controls the operation of the microprocessor. (ROM)
2. Control address register : is used to hold the address of the next micro instruction to be executed.
3. Sequencer : It is responsible for generating the sequence of addresses that are used to access the microprogram stored in the control memory.
4. Control bus : is used to transmit control signals between the control unit and the other components of the microprocessor.
5. Microinstruction register : used to hold the microinst. that is currently being executed.
6. Decoder : used to Decode the microinstr. in the MIR.
7. Data bus : Used to transmit data b/w microprocessor and memory or I/O devices.

The operations of a microprogrammed control unit involves the following steps :

1. The sequencer generates the address of the next microinstruction to be executed and stores it in the CAR.
2. The Control Memory is accessed using the address in the CAR, and the corresponding microinstruction is loaded into the MIR.
3. The decoder decodes the microinstruction in the MIR and generates the appropriate control signals to execute the corresponding operation in the microprocessor.
4. The microprocessor performs the operation specified by the microinstruction, and the next microinstruction is fetched and executed.

Q. Define micro operation and micro code.

Ans Micro-operation:
also known as a micro-instruction, is a basic operation performed by a digital computer's control unit at the hardware level. Micro-operations are the fundamental building blocks used by the control unit to execute instructions and manage data movement within computer.

Micro-Code:
Micro code is a set of microinstructions that defines the control logic of a microprogrammed control unit. Micro code is a low-level software that controls the operation of a computer's central processing unit (CPU) at a hardware level.

Q. write short note on RISC.

Ans RISC processors use a small set of simple instructions that can be executed quickly.

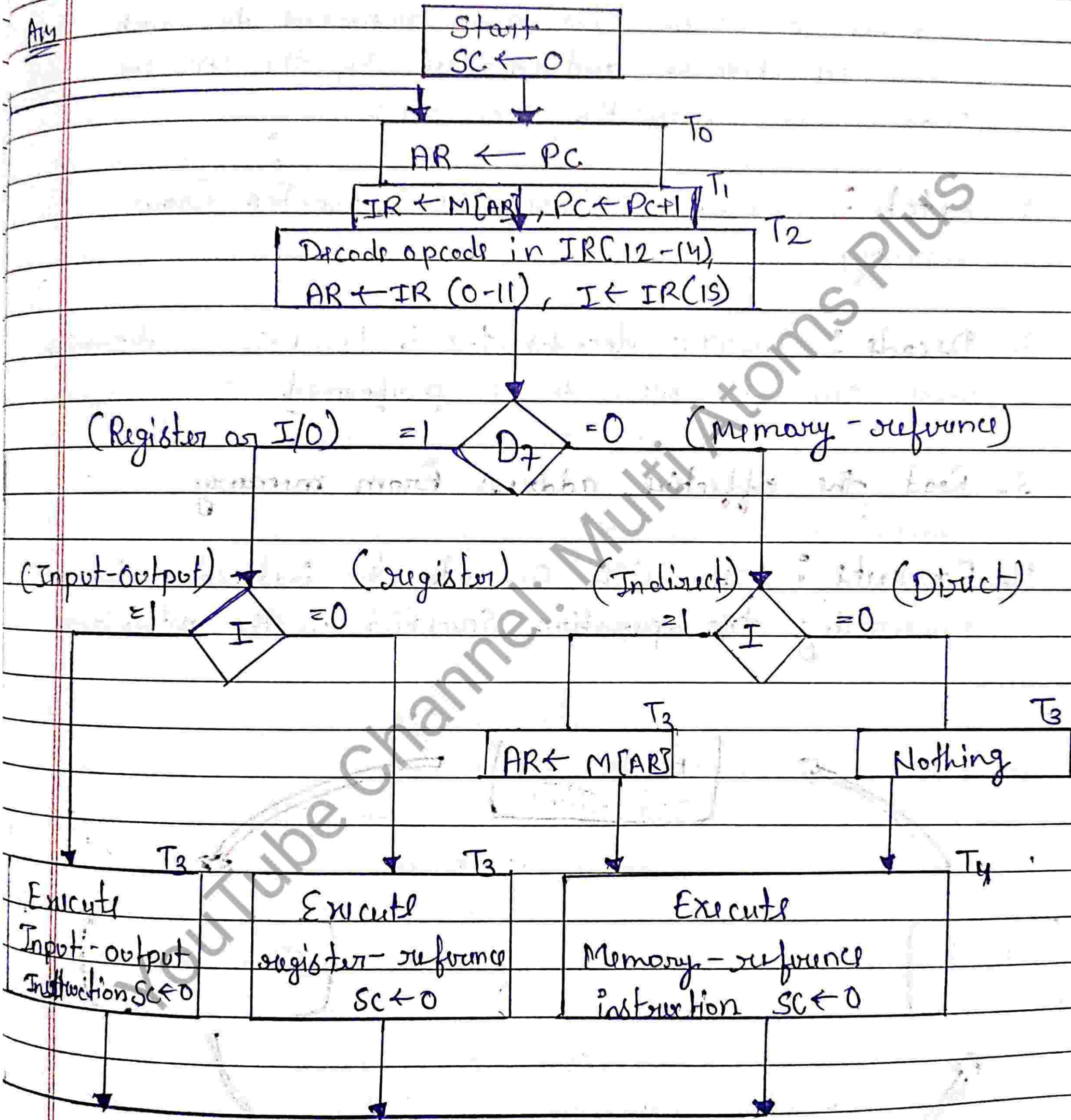
→ Each instruction can be executed in a single clock cycle.

→ RISC processors use a Load / Store architecture.

→ RISC processors often use highly pipelining to improve speed.

Q. Draw the flowchart for instruction cycle with neat diagram and explain.

Ans



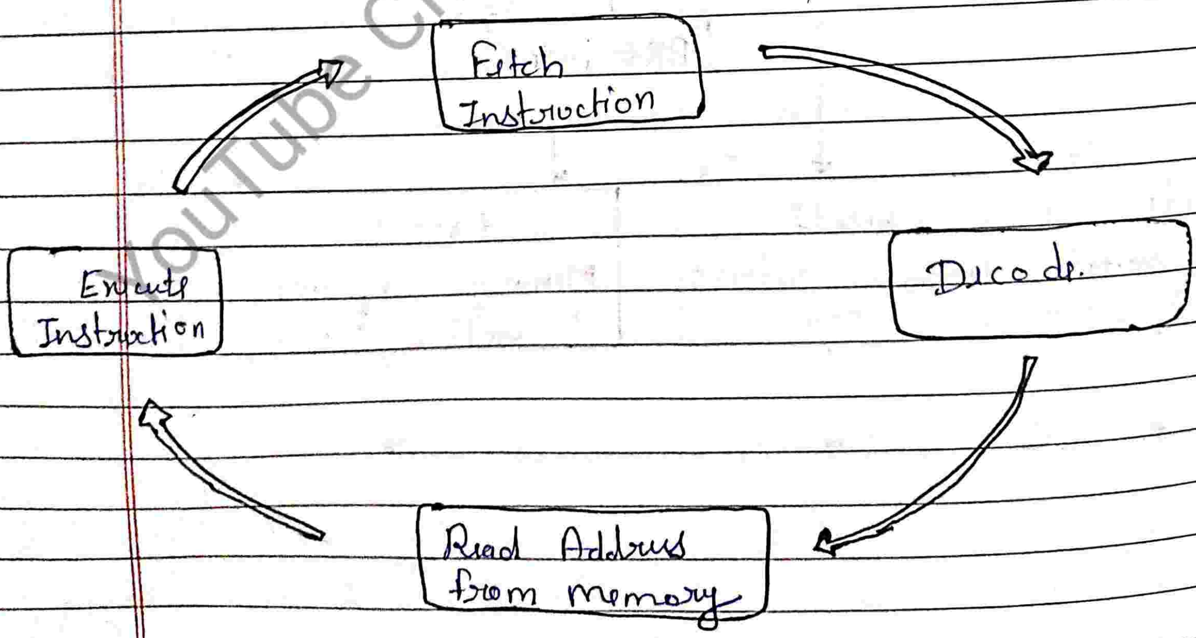
The instruction cycle is the basic operation performed by a computer's Central processing unit (CPU) to execute instructions. It consists of a sequence of steps that are repeated for each instruction fetched and executed by the CPU. The steps in the instruction cycle are:

1. Fetch : The CPU fetches the instruction from memory.

2. Decode : The CPU decodes the instruction to determine what operation needs to be performed.

3. Read the effective address from memory.

4. Execute : The CPU executes the instruction by performing the operation specified in the instruction.



Q. what is a micro program sequencer? with block diagram, explain the working of micro program sequencer.

Ans

Microprogram Sequencer

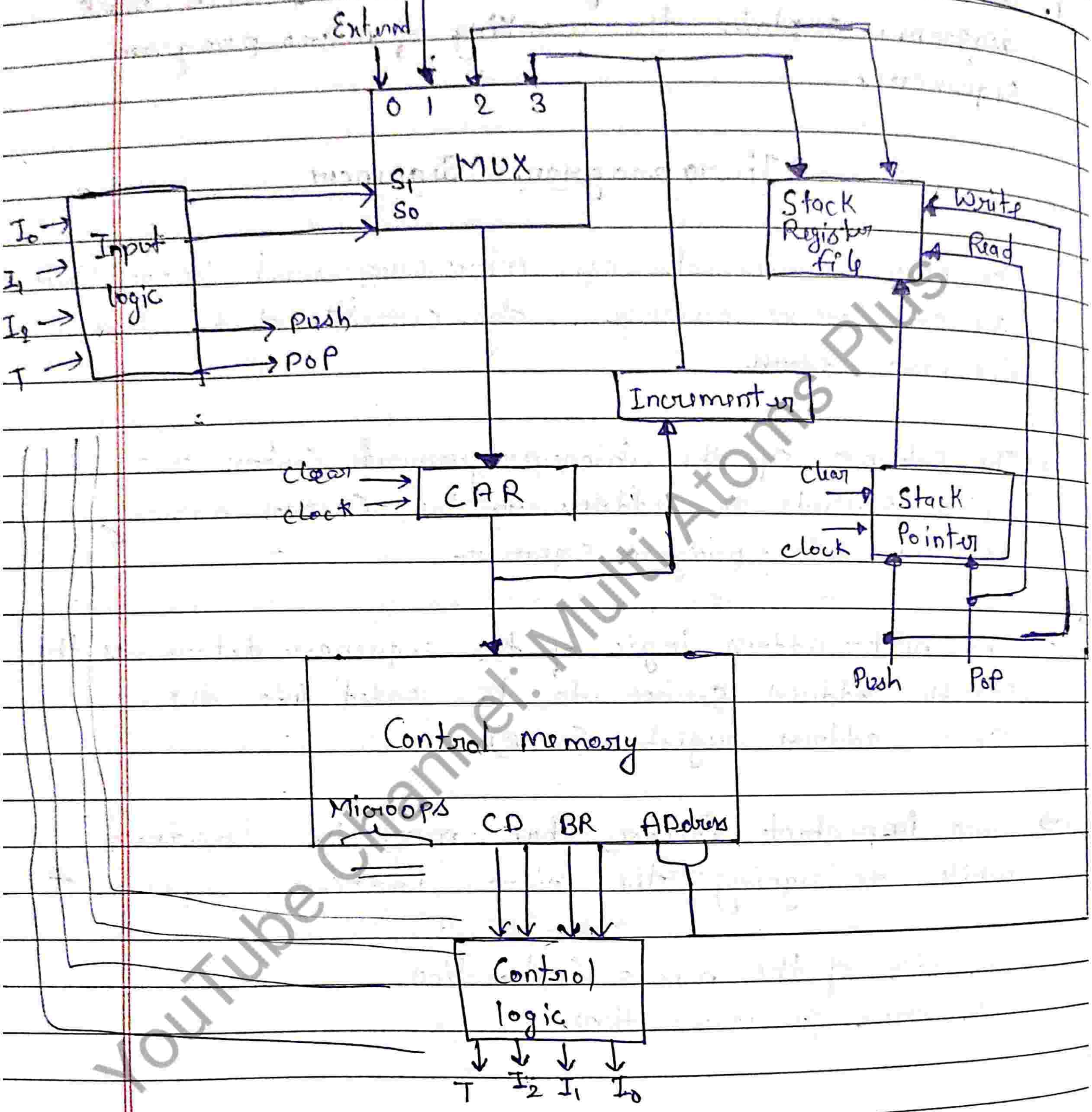
The basic components of micro programmed control unit are the control memory & the circuits that select the next address.

→ The subunit of the microprogrammed control unit which presents an address to the control memory is called micro program sequencer.

→ The next-address logic of the sequencer determines the specific address source to be loaded into the control address register (CAR).

→ Two important factor that must be considered while desisigning the micro instruction sequencer -

i) The size of the micro instruction.
ii) The address generation time.



→ The purpose of micro program Sequencer is to present an address to the control memory so that a micro instruction may be read & executed.

→ Here multiplexer selects an address from 4 sources & routes it into a Control address register.

→ The output from CAR Provides the address for the control memory.

→ The Contents of CAR are incremented & applied to the multiplexer & to the stack register file.

→ The register selected in the stack is determined by the stack pointer.

→ Input I_2, I_1, I_0, & T (test) derived from CD & BR fields of microinstructions specify the operation for the sequencer.

They specify the input source to the multiplexer also generate push & pop signals required for stack operation.

Typical Sequencer operations are:

- Increment
- Branch
- Push or PoP
- Call & return

~~Jump~~ Function Table of Micro program Sequencer

	I_2	I_1	I_0	T	S_1	S_0	Operation	Description
1.	X	0	0	X	0	0	CAR ← EXA	External address.
2.	1	0	1	1	0	1	CAR ← BRA SR ← CAR+1	Branch to Subroutine Save the next add.
3.	0	0	1	1	0	1	CAR ← BR	transfer branch add.
4.	X	1	0	X	1	0	CAR ← SR	Tranfrom from stack Register.
5.	0	1	1	0	1	1	CAR ← CAR+1	Increment add.

Q. Differentiate between hardwired and micro programmed control unit. Explain each component of hardwired control unit organization.

S.No.	Characteristics	Hardwired Control	Micro-programmed Control
1.)	Speed	Fast	Slow
2)	Implementation	Hardware	Software.
3)	flexibility	Not flexible	flexible.
4.)	Ability to handle large instruction set	Difficult	Easier
5).	Ability to support OS and diagonostic features	Very difficult	Easy
6)	Design process.	Difficult for more operation	Easy.
7)	Memory	Not used	RAM or ROM.
8).	chip area	less area	more area
9.)	Used in	RISC processor	CISC processor
10)	Output generation	Basis of input signal	Basic of control line.

The Component of Hardwired Control Unit.

1. Instruction register (IR) = It holds the current instruction being executed.

2. Instruction decoder : It decodes the instruction and generates control signals for the rest of the processor.

3. Control logic :- ~~It decodes the instruction and generates control signals for the rest of the processor.~~
→ It generates the control signals based on the decoded instruction and the state of the processor.

4. ALU Control : It generates control signals for the arithmetic and logic unit (ALU).

5. Register file control :- It generates control signals for the register file.

6. Clock :- It provides timing signals for the operation of the control unit.

Q. Differentiate between horizontal and vertical microprogramming.

Horizontal	Vertical
→ That uses no encoding at all.	→ The control bits are encoded in it.
→ It support longer control word.	→ It supports shorter control word.
→ It is faster	→ It is slower.
→ more flexible	→ less flexible.
→ No additional hardware is required	→ Additional hardware required in the form of decoder.

Q. What are the different types of instruction formats?

Ans Zero - instruction format
One - instruction format
Two - instruction format
three - instruction format
fixed - instruction format
Variable - instruction format

Q. Explain with neat diagram, the address selection for control memory.

Ans pag no :- 63 micro program Sequencer.

Q. write a program to evaluate arithmatic expression using stack organized Computer with 0-address instructions.

$$X = (A - B) * (((C - D * E) / F) / G)$$

Ans

1.	PUSH	A	TOS ← A
2.	PUSH	B	TOS ← B
3.	SUB		TOS ← (A − B)
4.	PUSH	C	TOS ← C
5.	PUSH	D	TOS ← D
6.	PUSH	E	TOS ← E
7.	MUL		TOS ← (D * E)
8.	SUB		TOS ← (C − (D * E)).
9.	PUSH	F	TOS ← F.
10.	DIV		TOS ← (C − (D * E)) / F).
11.	PUSH	G	TOS ← G.
12.	DIV		TOS ← (C − (D * E)) / F) / G.

13. MUL (A-B)*((C-(D*E)/F)/G).
14. POP M[X] ← TOS.

Q. List the differences between hardwired and micro programmed control in tabular format.

→ page no. 66.

Q. write the sequences of control steps for the following instruction for single bus architecture.

R1 ← R2 * (R3).

Ans 1. Fetch the instruction from memory and decode the opcode and operands.
2. Fetch the content of R2 from the register file and place it on the bus.
3. Fetch the contents of R3 from the register file and place it on the bus.
4. Send the signal to the ALU to initiate the multiption operation.
5. wait for the ALU to complete the multiplication operation.
6. Fetch the result of the multiplication from the output of the ALU on the bus.
7. Write the result to R1 in the register file.